F7AB - Actuarial and Financial Mathematics
Tutorial 7 – IRR, DPP and profits

Question 1 You invest £1 000 in a project and receive back £218.00 at the end of the first year of the project, £356.43 at the end of the second year and £647.51 at the end of the third year.

1. Write down an equation for the net present value (NPV) from the project and evaluate it at an interest rate of: (a) 8% p.a. effective; and (b) 9% p.a. effective.

2. Calculate the internal rate of return (IRR) on the project.

Question 2 You borrow £1 000 from your bank at an effective rate of interest of 6% p.a. Using your borrowings, you invest in a project which pays £2 158.92 at the end of ten years.

1. Calculate your internal rate of return (IRR).

You repay the bank at the end of ten years (repaying the loan and the interest accumulated over the ten years) out of the payment from the project.

2. Calculate your profit at the end of the ten years.

Question 3 You borrow £1 000 from the bank at an effective rate of 6% p.a. and invest it in a project. The project provides you with a payment of £400 at the end of the first year. From this payment you repay the interest you are owing to the bank and use the balance of the payment to repay as much as you can of the loan from the bank. At the end of the second year you receive £800 from the project. From this payment, you repay both the interest you are owing to the bank and the outstanding loan to the bank. What you have left over, i.e. the balance, is profit for you.

1. Calculate the profit.

2. Calculate your internal rate of return (IRR) on the project.

Question 4 You borrow £1 000 000 for a company that will make profits of £200 000 at the end of each of the next 10 years. The lender charges interest at 7% per annum effective. You will use the profits to repay the loan as soon as possible, and then any further profits can be invested to earn interest at 6% per annum effective.

1. Calculate the Discounted Payback Period (DPP).

2. Calculate the accumulated profit at the end of 10 years.

Question 5 You borrow £100 million to set up a factory which will make profits of £15 million at the end of each of the next 10 years. You agree to repay the loan by annual level repayments over the 10 years of the venture, calculated at an interest rate of 5% per annum effective. The net profits are invested to earn 3% per annum effective until the end of 10 years.

1. Calculate your accumulated profit at the end of 10 years.

2. Calculate how much more accumulated profit you would have made if you had used all of the profits in the early years to repay the loan as fast as possible.

Question 6 You wish to borrow £10 million for a project, which will return profits of £1.5 million at the end of each of the first 5 years, and £2 million at the end of each of the next five years. You want to maximise your accumulated profits at the end of the project.

You have three possible sources of loan capital.

(I) Borrow it from bank A, which charges interest at 8% per annum, and use the net profits from the venture to repay the loan as soon as possible.

(II) Borrow it from bank B, which charges interest at 7.5% per annum, and repay it by level annual repayments over the ten years.

(III) Issue a bond with term 10 years and a coupon of 7% per annum, payable annually. The bond will be issued at par and redeemed at 105%.

Net profits can be invested to accumulate at 5% per annum effective. What method of financing should you choose?

Question 7 You borrow £100 million and build a widget factory. It will return profits at the end of each of the next 10 years, starting with £14 million in the first year and subsequently increasing by 5% compound each year.

You borrow the money from a bank which charges interest of 7% per annum on loans, and you will use the profits to repay the loan as quickly as possible. Further profits can then be invested at 5% per annum effective.

Calculate your accumulated profit at the end of 10 years.

Answers: Q1.1(a). 21.447; Q1.1(b). -0.003; Q1.2. 9%; Q2.1. 8.000%; Q2.2. £368.07; Q3.1. £100.40; Q3.2. 11.652%; Q4.1. 7 years; Q4.2. £785 623; Q5.1. £23.50 million; Q5.2. £2.073 million; Q6. II; Q7. £39.721 million.